

HAMILTON–BOLTZMANN COTANGENT GEOMETRY FROM DETERMINISTIC HARD-SPHERE COLLISIONS

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ABSTRACT. The density–actual-contact action is placed in one weighted strict Radon duality, with separate local analytic and global convex source domains. The complete cotangent is the Hausdorff quotient by $(r, -\Delta r)$, and arbitrary bounded-source exhaustion recovers the full entropy action and detects singular current. Local covariance is strictly positive exactly modulo the balance gauge, endpoint coboundaries, and prepared constraints. Exact finite-volume centring and time-localized connected cumulants yield a process-level joint Gaussian limit with all density–contact cross covariances and without an assumed point-process compensator.

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1. ROUND-SIX POSITIVE CLOSURE: TWO-DOMAIN COTANGENTS AND EXACT-CENTRED GAUSSIAN FIELDS

The local analytic source chart used for pressure derivatives is separated from the global convex source domain used for Fenchel duality. The latter is the union of all bounded compactly supported density/contact cylinders,

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so it can be scaled and truncated without leaving the proved B2 pressure domain. The process Gaussian limit is obtained from time-localized connected cumulants and conditional block estimates, not from an assumed point-process compensator. Every microscopic field is centred at its exact finite-volume expectation.

1.1. Weighted primal and cotangent spaces. Put

$$w(v) = 1 + |v|^2, \quad w_c(v, v_*) = 1 + |v|^2 + |v_*|^2.$$

Let $\mathcal{X}_T$ be the density-path space with the topology generated by weighted bounded-Lipschitz tests and one Maxwellian Lyapunov moment inherited from B2. Let $\mathfrak{C}_T$ be the exchange-quotiented actual-contact space, and let $\mathcal{M}_{w_c}^+(\mathfrak{C}_T)$ be the positive Radon measures with finite w_c -moment.

Let $C_{w_c,0}(\mathfrak{C}_T)$ be the continuous functions z such that z/w_c vanishes at weighted infinity, equipped with the weighted strict topology β_{w_c} . Let $\mathcal{Y}_p$ be the completion of smooth time-dependent density tests for

$$\begin{aligned} \|p\|_{\text{graph}} = & \|p_0/w\|_{\infty} + \|p_T/w\|_{\infty} + \|((\partial_t + v \cdot \nabla_x)p)/w\|_{\infty} \\ & + \|\Delta p/w_c\|_{\infty} + \text{the corresponding weighted strict moduli.} \end{aligned}$$

Then $p \mapsto \Delta p$ is continuous from $\mathcal{Y}_p$ to $C_{w_c,0}$.

A pair (f, Γ) is balanced if

$$\langle p_T, f_T \rangle - \langle p_0, f_0 \rangle - \int_0^T \langle (\partial_t + v \cdot \nabla_x)p, f_t \rangle dt = \int \Delta p d\Gamma$$

for every $p \in \mathcal{Y}_p$.

1.2. Local analytic and global convex source domains. For a regular Maxwellian-biased phase with tail exponent $\beta_* > 0$, set

$$\mathcal{U}_{\eta} = \{(p, \psi) : \|\Delta p + \psi\|_{w_c} < \eta\}, \quad 0 < \eta < \beta_*/8.$$

On $\mathcal{U}_{\eta}$, the exponential Hamiltonian is finite and the B2 pressure is analytic.

For global duality, let $\mathcal{D}_{\text{glob}}$ be the union, over all $R, M < \infty$, of smooth density tests and continuous contact tests supported in $w, w_c \leq M$ and bounded in absolute value by R . B2 proves the finite-horizon pressure on every such bounded real source ball. Define its lower-semicontinuous extension to a weighted source by monotone truncation:

$$\overline{Q}(p, \psi) = \sup_M \lim_{R \rightarrow \infty} Q(p^{M,R}, \psi^{M,R}),$$

whenever the right-hand side is proper. The local analytic pressure and the global extension agree on $\mathcal{U}_{\eta} \cap \mathcal{D}_{\text{glob}}$.

Lemma 1.1 (Strict Radon dual). *The continuous dual of $(C_{w_c,0}, \beta_{w_c})$ is the space of countably additive signed Radon measures $\mathcal{M}_{w_c}$, under integration. No finitely additive functional occurs.*

Proof. Restrict a continuous functional to every compact sublevel $\{w_c \leq M\}$. There the strict topology agrees with uniform convergence on bounded sets, so Riesz representation gives a compatible Radon measure. Continuity on weighted bounded sets makes the weighted total variations uniformly summable; the measures paste to one member of $\mathcal{M}_{w_c}$. Conversely, weighted tightness and dominated convergence make integration by any such Radon measure strictly continuous. $\square$

1.3. The exact gauge complex. Define

$$\mathfrak{G}r = (r, -\Delta r), \quad \mathfrak{D}(p, \psi) = \Delta p + \psi.$$

Theorem 1.2 (Closed balance-adjoint gauge). *The sequence*

$$0 \longrightarrow \mathcal{Y}_p \xrightarrow{\mathfrak{G}} \mathcal{Y}_p \times C_{w_c,0} \xrightarrow{\mathfrak{D}} C_{w_c,0} \longrightarrow 0$$

is exact and has closed first range. Hence

$$(\mathcal{Y}_p \times C_{w_c,0}) / \text{Ran } \mathfrak{G} \cong C_{w_c,0}$$

as Hausdorff locally convex spaces. On every balanced pair the dual action depends only on $z = \Delta p + \psi$. Collision invariants are included as normalized gauge directions but do not exhaust the gauge.

Proof. Surjectivity follows from $z = \mathfrak{D}(0, z)$. If $\mathfrak{D}(p, \psi) = 0$, then $(p, \psi) = \mathfrak{G}p$, proving exactness. The range of $\mathfrak{G}$ is the graph of the continuous map $r \mapsto -\Delta r$, hence is closed. On a balanced pair, the endpoint and transport terms equal $\int \Delta p d\Gamma$, so the action contains p, ψ only through their sum. Fixing the finite-dimensional collision invariants selects a representative without changing the quotient class. $\square$

1.4. Global integral Fenchel duality. For a density path f , let

$$dA_f = \frac{1}{2} f f_* ((v - v_*) \cdot \omega)_+ dt dx dv dv_* d\omega,$$

and put

$$\mathfrak{J}_f(\Gamma) = \begin{cases} \int \ell(d\Gamma/dA_f) dA_f, & \Gamma \ll A_f, \\ +\infty, & \text{otherwise,} \end{cases} \quad \ell(q) = q \log q - q + 1.$$

For a bounded global test define

$$\begin{aligned} \mathfrak{L}_{f,\Gamma}(p, \psi) &= \langle p_T, f_T \rangle - \langle p_0, f_0 \rangle - \int \langle (\partial_t + v \cdot \nabla_x) p, f \rangle dt \\ &\quad + \int \psi d\Gamma - \int (e^{\Delta p + \psi} - 1) dA_f. \end{aligned}$$

Theorem 1.3 (Two-domain path-space duality). *For every finite-moment pair (f, Γ) ,*

$$\sup_{(p,\psi) \in \mathcal{D}_{\text{glob}}} \mathfrak{L}_{f,\Gamma}(p, \psi) = \begin{cases} \mathfrak{J}_f(\Gamma), & (f, \Gamma) \text{ balanced,} \\ +\infty, & \text{otherwise.} \end{cases}$$

For a regular pair with $0 < c \leq d\Gamma/dA_f \leq C < \infty$, the optimizer belongs to a local analytic chart and its unique quotient representative is

$$z = \log(d\Gamma/dA_f).$$

The global theorem does not assert that arbitrarily large truncations remain inside one analytic chart.

Proof. If balance fails, a compactly supported smooth density test p has nonzero balance defect. Set $\psi = -\Delta p$; both are bounded on their compact velocity support and belong to $\mathcal{D}_{\text{glob}}$. Scaling the pair by any finite constant leaves the exponential argument zero and sends the linear defect to infinity.

On the balanced set, the action reduces to

$$\int z d\Gamma - \int (e^z - 1) dA_f.$$

For the absolutely continuous part, take bounded truncations of $\log(d\Gamma/dA_f)$, followed by compact velocity cutoffs. Monotone convergence gives $\int \ell(d\Gamma/dA_f) dA_f$. If Γ has a singular part, choose a compact set K charged by it and open neighborhoods U_n with $A_f(U_n) \leq e^{-2n}$. A continuous Urysohn test equal to n on K and supported in U_n gives a linear term of order $n\Gamma(K)$ and an exponential cost at most e^{-n} . Hence the supremum is infinite. Every test used is bounded for fixed n , although the sequence ranges over increasing source balls. Strict convexity of $e^z - 1$ gives the regular optimizer. Its bounded logarithm lies in a local chart after choosing the chart around that regular phase. $\square$

1.5. The covariance cohomology. Let Q_Θ be the local analytic prepared pressure at a regular biased source Θ , and let $\Sigma_\Theta = D^2 Q_\Theta$. Gauge directions are identically zero along every microscopic balanced path and therefore lie in its kernel.

Theorem 1.4 (Complete local covariance kernel). *On a regular analytic source chart, the kernel of the speed-normalized pressure covariance consists exactly of the closed balance gauge, the fixed initial constraints removed by B1, and endpoint/time coboundaries represented in $\mathcal{Y}_p$. Equivalently, the Hessian is strictly positive on every finite-dimensional family of independent quotient classes.*

Proof. The B2 lower-bound tilt identifies the local Legendre transform with the relative-entropy action. Its second variation at a regular pair is

$$\delta^2 I(\dot{f}, \dot{\Gamma}) = \delta^2 I_0(\dot{f}_0) + \int \frac{\dot{q}^2}{q} dA_f + \text{the quadratic variation of } A_f \text{ in } \dot{f},$$

restricted to the linearized balance. B2's conservative regularization gives a dense set of smooth balanced tangent directions. If a source quotient class is nonzero, Hahn–Banach supplies one such tangent with nonzero pairing; the positive quadratic action makes the local support function strictly curved in

that direction. Legendre Hessian inversion therefore gives positive pressure variance.

Conversely, a zero pressure variance pairs to zero with every finite-action balanced tangent. Density of the regular tangents and the closed graph gauge of Theorem 1.2 imply that the source lies in the balance adjoint range, plus an endpoint coboundary. B1 removes exactly the finite initial constraint span by its Schur complement. This proves the stated kernel without asserting that a zero-variance observable is pointwise constant. $\square$

1.6. Exact finite-volume centring and localized cumulants. Under the finite-volume biased law, define

$$\bar{\pi}_\varepsilon(t) = \mathbb{E}_{\Theta, \varepsilon} \pi_t^\varepsilon, \quad \bar{\Gamma}_\varepsilon = \mathbb{E}_{\Theta, \varepsilon} \Gamma_T^\varepsilon,$$

and the exactly centred fields

$$\zeta^\varepsilon = \sqrt{\mu_\varepsilon}(\pi^\varepsilon - \bar{\pi}_\varepsilon), \quad \Xi^\varepsilon = \sqrt{\mu_\varepsilon}(\Gamma^\varepsilon - \bar{\Gamma}_\varepsilon).$$

For an interval $I \subset [0, T]$, write $Y_I^\varepsilon(\phi, \psi)$ for the corresponding density increment plus actual-contact test, centred at its exact finite expectation.

Lemma 1.5 (Time-localized cumulant bounds). *For every bounded finite family of density/contact tests and every $k \geq 2$,*

$$|\kappa_k(Y_I^\varepsilon)| \leq C_k |I| \mu_\varepsilon^{1-k/2} + C_k \mu_\varepsilon^{-k/2}.$$

Conditionally on a stopping time τ and the complete trace hierarchy at τ , the same estimate holds for $I = [\tau, \tau+h]$ with $|I| = h$, uniformly on exponentially compact trace balls. In particular,

$$\mathbb{E}|Y_{[s,t]}^\varepsilon|^4 \leq C|t-s|^2 + C|t-s|/\mu_\varepsilon + C\mu_\varepsilon^{-2}.$$

Proof. Insert a source supported in I into the B2 trace-class connected pressure. A connected contribution meeting that source has one distinguished time in I , giving the factor $|I|$; a possible endpoint atom gives the smaller $\mu_\varepsilon^{-k/2}$ term. Normal convergence on a complex neighborhood bounds every derivative. Scaling by $\sqrt{\mu_\varepsilon}$ gives the stated cumulants. After a stopping time, the exact Markov property holds on the complete finite-volume law/trace state, not on density alone. B2's conditional trace estimate places every reached state in the same compact trace ball, so the same local source calculation applies. The fourth-moment identity $\mathbb{E}Y^4 = 3\kappa_2(Y)^2 + \kappa_4(Y)$ gives the final estimate. $\square$

1.7. The full joint Gaussian process. Let $(f, \Gamma = qA_f)$ be the limiting regular biased pair. The linearized balance propagator is denoted by $U(t, s)$. For density/contact tests (ϕ, ψ) and (χ, η) , define the instantaneous collision covariance

$$\mathcal{C}_t((\phi, \psi), (\chi, \eta)) = \int q_t(\Delta\phi + \psi)(\Delta\chi + \eta) dA_{f_t}.$$

This one formula includes all density–density, density–contact, and contact–contact cross terms.

Theorem 1.6 (Exact-centred density–actual-contact Gaussian limit). *The fields $(\zeta^\varepsilon, \Xi^\varepsilon)$, centred at their exact finite-volume expectations, converge in the weighted nuclear distribution path space to the unique centred Gaussian solution of the linearized biased balance with initial covariance equal to the B1 constrained Schur complement and dynamic covariance*

$$\int_0^T C_t(U(T, t)^*(\phi, \psi), U(T, t)^*(\chi, \eta)) dt.$$

Every finite-dimensional covariance is the Hessian of the limiting prepared pressure. No predictable Poisson compensator for the deterministic contact measure is assumed.

Proof. For finitely many time-localized tests, analytic convergence of the B2 pressure gives convergence of second cumulants. Lemma 1.5 makes all cumulants of order at least three vanish, hence gives the Gaussian finite-dimensional limit. Differentiating the limiting marked Hamilton–Jacobi equation twice yields the linearized propagator and the bilinear contact covariance displayed above; because the same actual contact carries both $\Delta\phi$ and ψ , the cross terms are included automatically.

For tightness, apply the conditional version of Lemma 1.5 to stopping-time increments. The $h^2 + h/\mu_\varepsilon$ estimate and the fact that individual empirical jumps are $O(\mu_\varepsilon^{-1/2})$ give Aldous tightness for every test coordinate. A countably Hilbert nuclear scale with Maxwellian weights and Mitoma’s theorem upgrade coordinate tightness to the distribution space. Every subsequential limit has the Gaussian finite-dimensional laws and the linearized balance, which determine it uniquely. B1 supplies the exact finite-volume conditioned initial covariance and its limiting Schur complement. $\square$

If one wishes to centre at the limiting means (f, Γ) , the deterministic bias $\sqrt{\mu_\varepsilon}((\bar{\pi}_\varepsilon, \bar{\Gamma}_\varepsilon) - (f, \Gamma))$ must be added explicitly; it is not silently discarded without a rate theorem.

Theorem 1.7 (Round-six B3 closure). *The density–contact action has a globally valid weighted-strict Radon dual and a separate local analytic pressure chart. Its cotangent is the Hausdorff balance-adjoint quotient, the local pressure covariance is nondegenerate exactly modulo gauge, endpoint coboundaries, and prepared constraints, and the exactly centred deterministic hard-sphere density/contact fields converge to the joint Gaussian process of Theorem 1.6.*

Proof. Combine Theorems 1.2, 1.3, 1.4, and 1.6, together with Lemma 1.1. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three

dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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